Multi-Agent Reinforcement Learning for Local Energy Market

gecad
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Baptiste TARET supervised by Pedro FARIA* & Roman LE GOFF LATIMIER

*GECAD, School of Engineering (ISEP) at the Polytechnic Institute of Porto (IPP)

Introduction

Energy Transition → Decentralized Renewable Generation → Complex, Dynamic Distribution Network → Peer-to-Peer Electricity Markets → Agents negotiate directly

Align individual actions with system needs

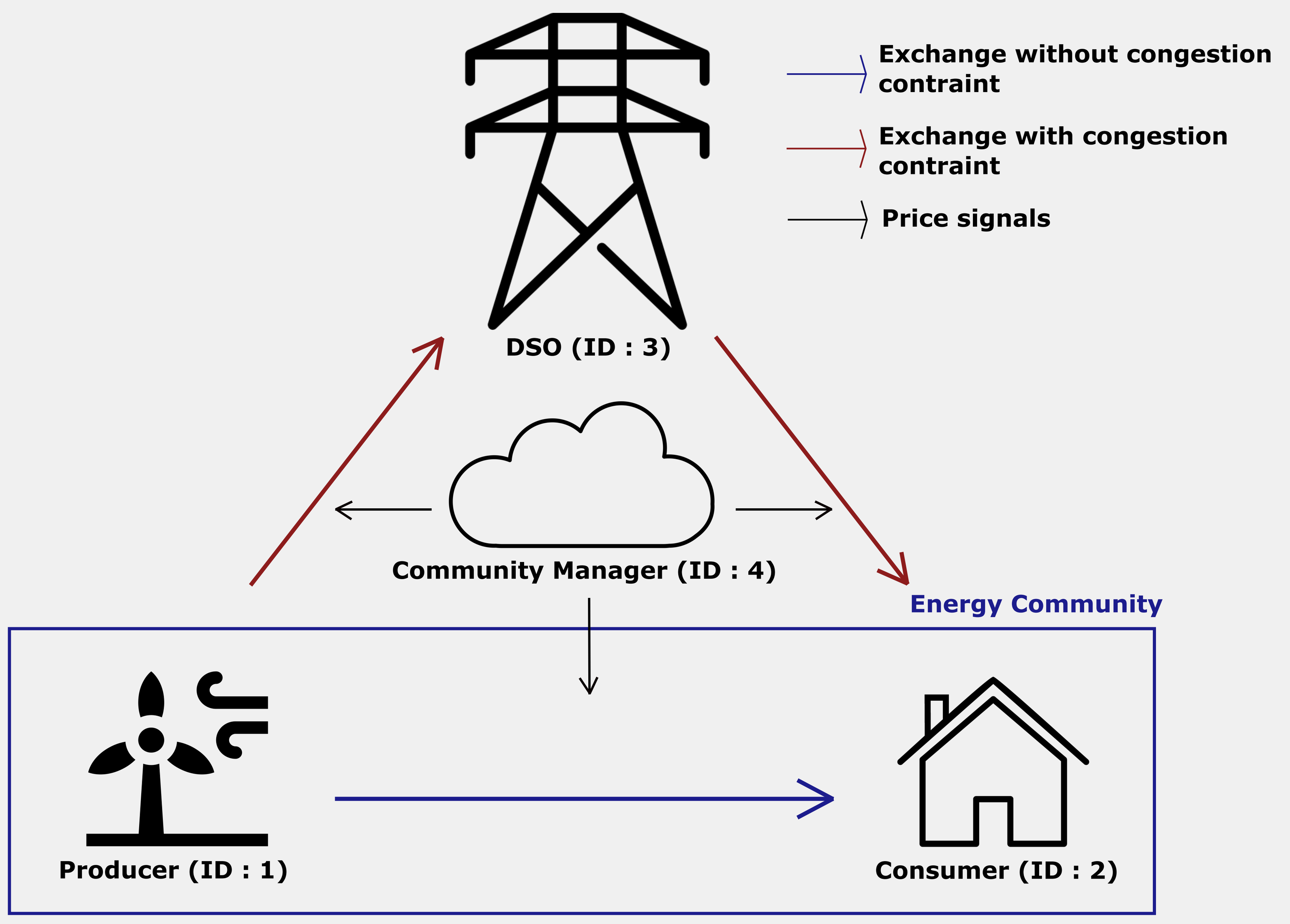


Illustration of the Simplified Local Energy Market (LEM) under Study

Objectives and method

LEM

- **Agents' desires ($\in \Omega$):** Convex quadratic function on power exchanges p_n
  - $f_n(p_n) = a_n p_n^2 + b_n p_n$ with $a_n > 0$, where agent n has ω_n neighbours
- **Function to minimize :** Convex quadratic function

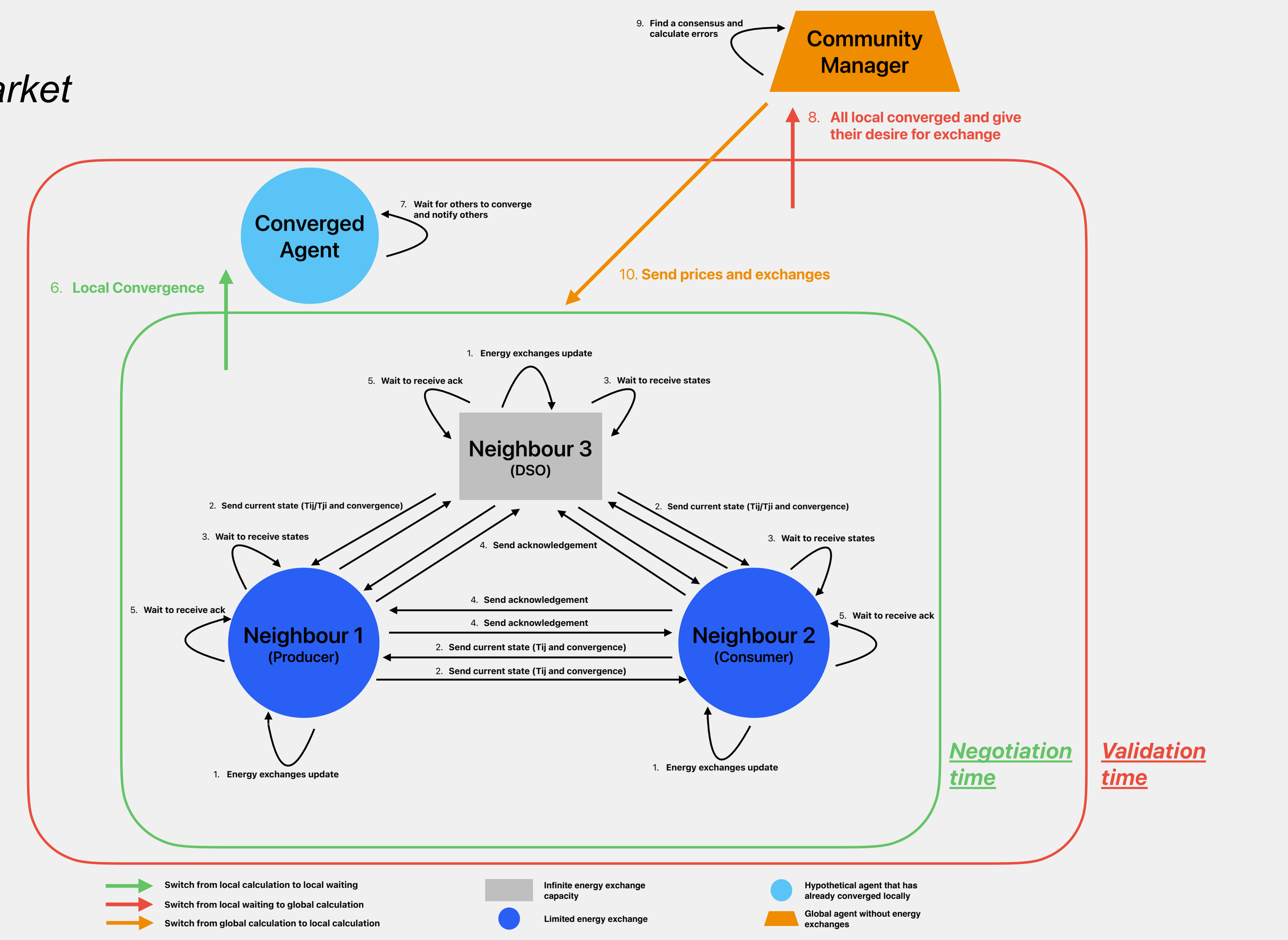
$$\sum_{n\in\Omega} f_n(p_n) + \sum_{m\in\omega_n} \gamma_{nm} t_{nm}$$

Price signals on trades from n to m

Trades, with $p_n = \sum_{m\in\omega_n} t_{nm}$

- **Constraints :** Maximum and minimum exchangeable power for each agent
- **Resolution Method (Constrained optimization problem):** Alterning direction method of multipliers (ADMM)
- **Communication between agents :** PEAK Python library based on SPADE

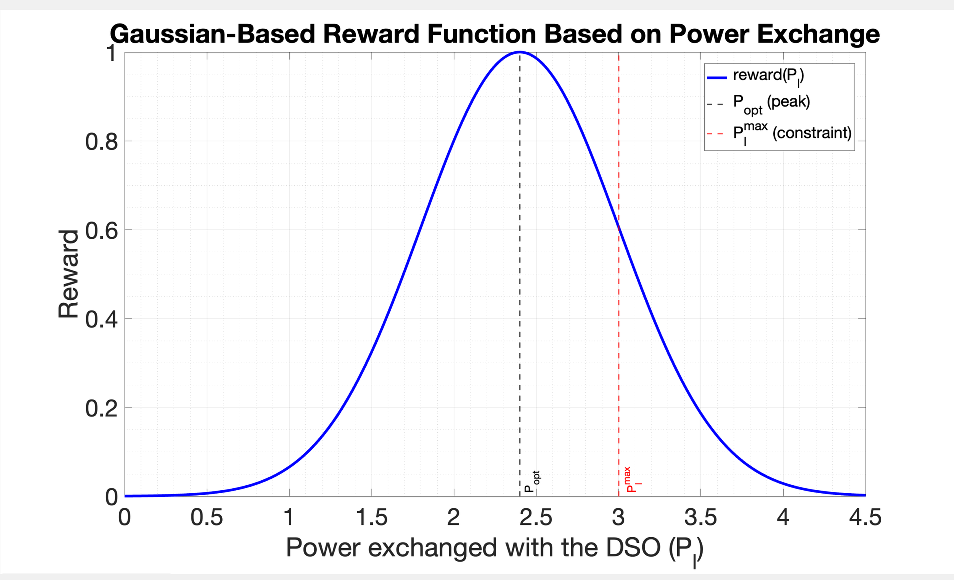
Multi-agent Market

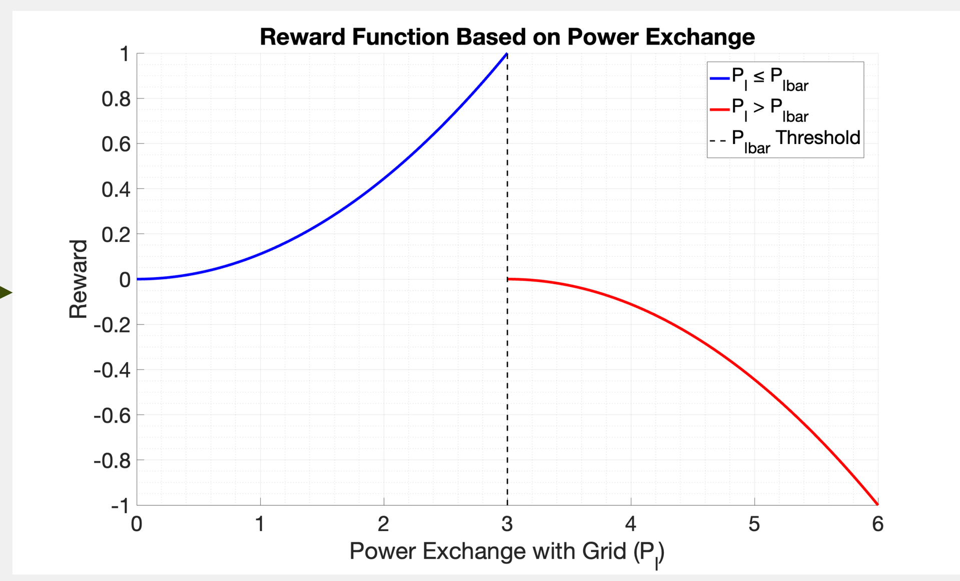


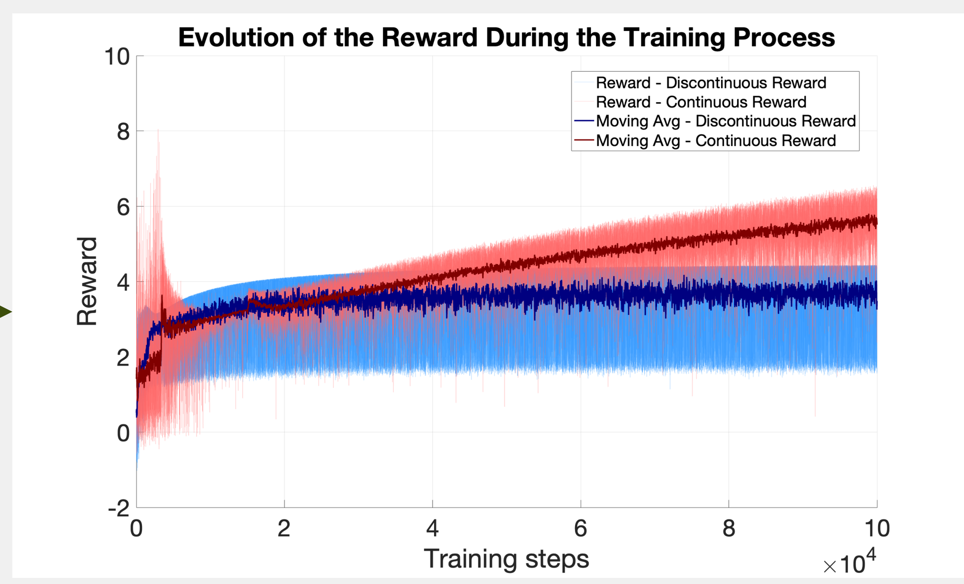
Communication Process of Multi-agent Energy Market

Results

Choosing the Most Effective Reward Function for Learning :

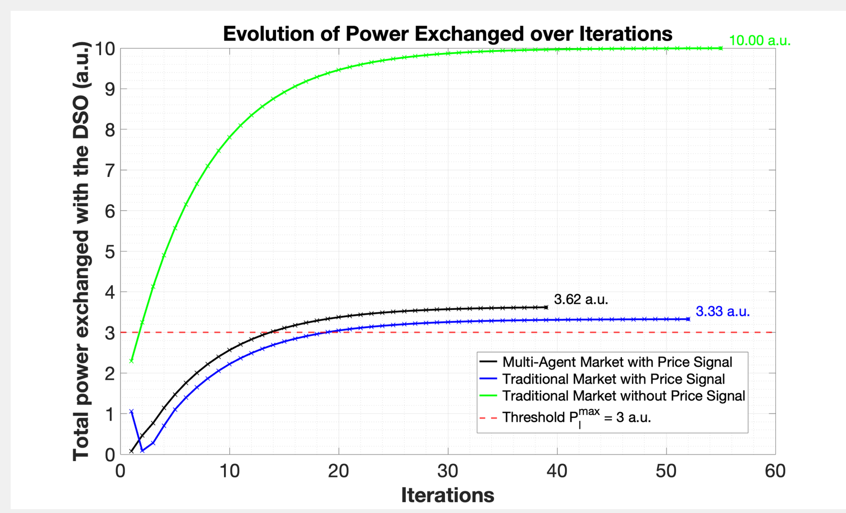




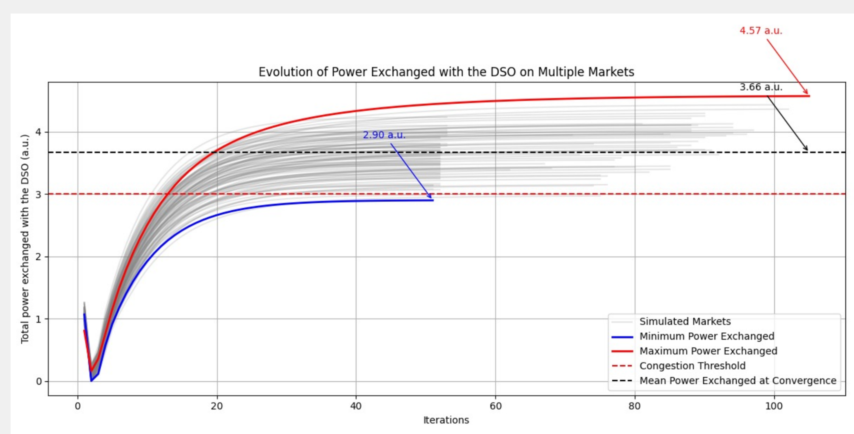


Continuous reward function sustain longer training and lead to **better learning**, while the maximum achievable reward remains unchanged

Practical Application of the Reward Design :



- **Overshoot reduced:** from **+233%** to **+11%**
- **Improvement:** ~**90%** better capacity compliance
- **Cause:** RL-based price signaling



- **Low variance** across simulated markets at convergence
- **Stable convergence** after ~20 iterations
- **Consistent final performance** despite different demand profiles

Outlooks

1. **Act only on prices related to exchanges with the DSO**
   - Reduced action space dimension

Stake :

- Evaluating if the agent can achieve similar behaviors with a reduced action space

2. Add a flexibility resource, such as a battery, to be controlled via reinforcement learning in order to minimize the price proposed by the DSO

- Simultaneously train two AIs: one to prevent congestion in DSO interactions, the other to control battery operation for minimizing DSO prices.